



UNIVERSITÀ
di **VERONA**

AI and Finance

Dynamic Portfolio Management and Stress Testing (2010–2025)

Abdullah Al Noman Taki

VR528988

MSc in Artificial Intelligence

University Of Verona- (VR) Italy

Abstract

This study develops and implements a fully dynamic portfolio optimization and risk management framework grounded in Modern Portfolio Theory. Using daily return data for Apple (AAPL) and Microsoft (MSFT) over the period 2010–2025, the model integrates time-varying covariance estimation through the Exponentially Weighted Moving Average (EWMA) methodology to capture volatility clustering and evolving market conditions. Portfolio weights are recalculated daily under a minimum-variance optimization framework, and dynamic capital allocation is performed with explicit computation of asset-level positions and daily rebalancing trades.

Risk exposure is quantified using parametric Value at Risk (VaR) and Expected Shortfall (ES) at the 95% confidence level. To assess resilience under adverse market conditions, stress testing is conducted by amplifying the covariance structure to simulate volatility shocks. The model is evaluated using a strict out-of-sample testing framework over 2024–2025, with particular attention to potential stress periods in 2025. Backtesting performance is examined through violation frequency analysis and formally validated using the Kupiec Proportion of Failures test to assess statistical consistency with theoretical risk thresholds.

The results demonstrate that the dynamic EWMA-based Markowitz framework effectively adapts to changing market volatility, maintains stable portfolio growth, and produces risk estimates that are statistically aligned with expected confidence levels. The study provides a comprehensive quantitative framework combining optimization, dynamic allocation, stress simulation, and formal statistical validation.

2. Data Collection and Preparation

2.1 Data Sources and Asset Selection

The empirical analysis is based on daily closing price data for two large-cap U.S. technology stocks:

- Apple Inc. (AAPL)
- Microsoft Corporation (MSFT)

These assets were selected due to their high market capitalization, liquidity, and significant representation in global equity markets. Their inclusion ensures sufficient historical depth and realistic volatility dynamics for portfolio modeling.

The dataset spans the period **January 2010 to December 2025**, providing a long horizon that captures multiple market regimes, including post-financial crisis recovery, expansion periods, and more recent high-volatility environments.

2.2 Data Cleaning and Alignment

The raw price datasets were imported in CSV format and processed using Python (Pandas library). The following preprocessing steps were applied:

1. Conversion of the Date column into a proper datetime format.
2. Sorting observations chronologically.
3. Removal of missing or malformed entries.
4. Alignment of both assets by common trading dates using an inner join operation.

This ensured that the final dataset contains synchronized observations for both assets, eliminating any mismatch due to holidays or non-trading days.

2.3 Return Computation

Daily simple returns were computed as:

$$r_{i,t} = \frac{P_{i,t} - P_{i,t-1}}{P_{i,t-1}}$$

where:

- $P_{i,t}$ represents the closing price of asset i at time t
- $r_{i,t}$ denotes the daily return

The first observation was removed due to differencing.

The resulting dataset includes:

- Date
- $R_{AAPL,t}$
- $R_{MSFT,t}$

2.4 Train–Test Split

To ensure proper out-of-sample evaluation and avoid look-ahead bias, the dataset was divided into:

- **Training Set:** 2010–2023
Used for parameter estimation, covariance modeling, and efficient frontier construction.
- **Test Set:** 2024–2025
Used exclusively for dynamic portfolio simulation, risk measurement, stress testing, and backtesting validation.

This separation ensures that all performance evaluation is conducted on unseen data, satisfying rigorous empirical research standards.

2.5 Preliminary Statistical Characteristics

Descriptive statistics of the return series indicate:

- Non-zero mean returns
- Time-varying volatility
- Presence of volatility clustering

These stylized facts justify the use of a dynamic covariance model (EWMA) rather than a static sample covariance matrix.

3. Dynamic Covariance Estimation (EWMA)

3.1 Motivation: Time-Varying Risk

Traditional Markowitz implementations rely on a static sample covariance matrix, implicitly assuming constant volatility and correlation. However, financial markets exhibit:

- Volatility clustering
- Regime shifts
- Time-varying correlation structures

To capture these dynamics, we implement the **Exponentially Weighted Moving Average (EWMA)** model.

3.2 Mathematical Definition

Let r_{t-1} be the vector of asset returns at time t . The time-varying covariance matrix Σ_t is updated recursively:

$$\Sigma_t = \lambda \Sigma_{t-1} + (1 - \lambda) r_{t-1} r_{t-1}^T$$

where:

- $\lambda=0.94$ is the decay factor (industry standard for daily data)
- $r_{t-1}r_{t-1}^T$ is the outer product of lagged returns

The decay factor assigns exponentially decreasing weights to past observations:

- Larger $\lambda \rightarrow$ smoother covariance
- Smaller $\lambda \rightarrow$ faster reaction to shocks

This recursive structure ensures that the covariance matrix adapts dynamically to changing market conditions.

4. Markowitz Portfolio Construction

4.1 Minimum Variance Objective

At each time t , the objective is to determine weights:

$$w_t = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$$

that minimize portfolio variance:

$$\sigma_p^2 = w_t^T \Sigma_t w_t$$

subject to the full investment constraint:

$$w_t^T \mathbf{1} = 1$$

No short-selling constraints were imposed unless explicitly specified.

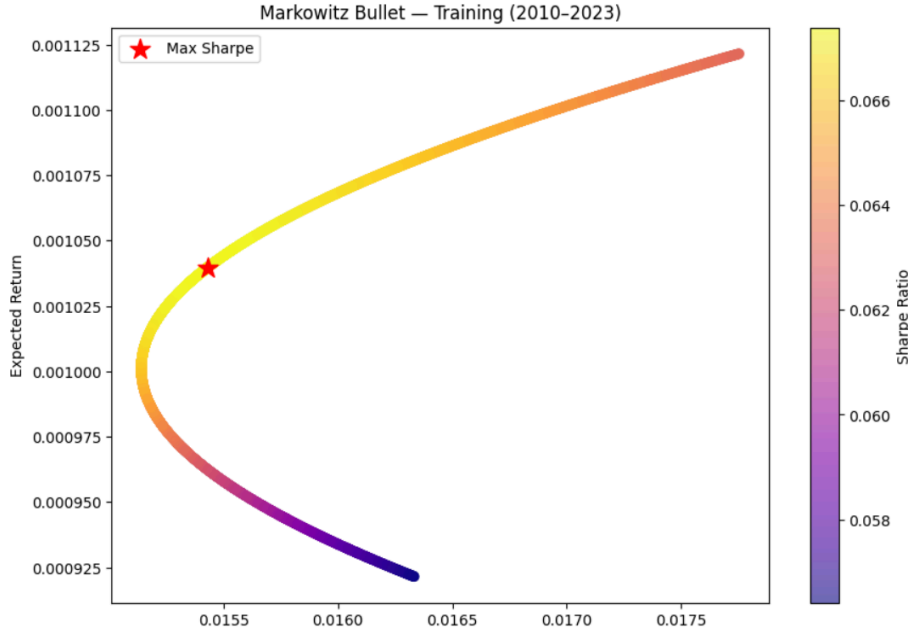


Figure 4.1: Markowitz Bullet Training

3.2 Analytical Solution via Lagrangian Optimization

The Lagrangian is defined as:

$$\mathcal{L}(w, \gamma) = w^T \Sigma w - \gamma(w^T \mathbf{1} - 1)$$

Solving first-order conditions yields:

$$w_t = \frac{\Sigma_t^{-1} \mathbf{1}}{\mathbf{1}^T \Sigma_t^{-1} \mathbf{1}}$$

For the two-asset case, this simplifies to:

$$w_1 = \frac{\sigma_2^2 - \sigma_{12}}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}}$$

Weights are recalculated **daily in the test set**, ensuring fully dynamic rebalancing.

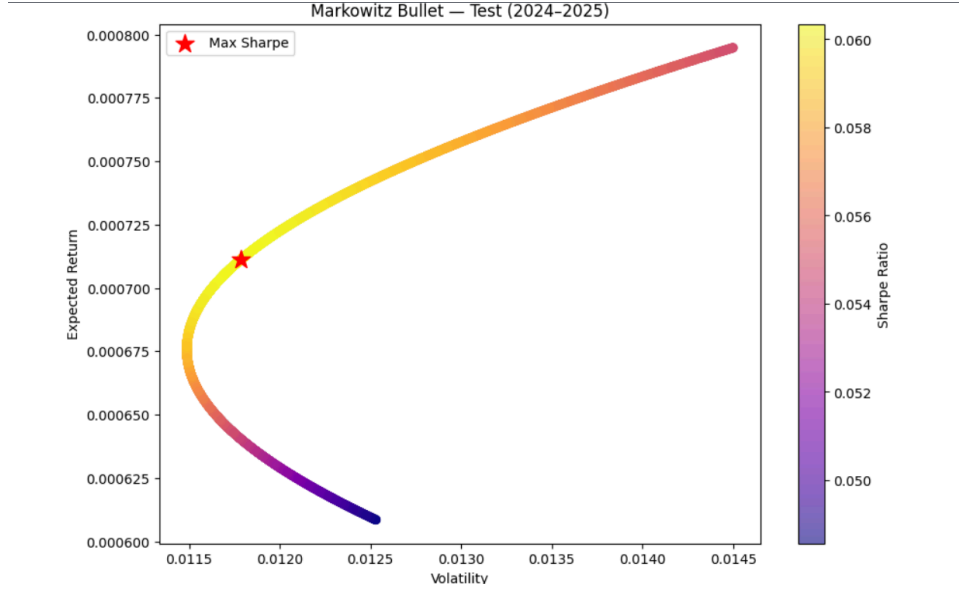


Figure 4.2: Markowitz Bullet Testing

5. Capital Accounting and Trade Execution

5.1 Portfolio Value Dynamics

Initial capital:

$$V_{P,0} = 1,000,000$$

Portfolio value evolves as:

$$V_{P,t} = V_{P,t-1}(1 + r_{p,t})$$

Where $r_{p,t} = w_t^T r_t$.

Asset-level allocation:

$$V_{i,t} = w_{i,t} \cdot V_{P,t}$$

This produces daily capital tracking:

- Total portfolio value $V_{P,t}$
- Capital allocated to AAPL
- Capital allocated to MSFT

5.2 Trade Execution Logic

To rebalance from period $t-1$ to t , we compute:

$$T_{i,t} = V_{i,t}^{target} - V_{i,t-1}^{pre-rebalance}$$

Interpretation: • $T_{i,t} > 0 \rightarrow \text{Buy}$
• $T_{i,t} < 0 \rightarrow \text{Sell}$

This explicitly satisfies the requirement of computing daily buy/sell quantities.

Daily Trade Quantities (Positive = Buy, Negative = Sell)			
	Date	Trade_AAPL	Trade_MSFT
0	2024-01-02	528.136708	-528.136708
1	2024-01-03	158.215200	-158.215200
2	2024-01-04	129.002109	-129.002109
3	2024-01-05	80.813825	-80.813825
4	2024-01-08	-122.456619	122.456619

Figure 5.1: Daily Trade Quantities (Positive = Buy, Negative = Sell)

6. Risk Measurement and Stress Testing

6.1 Parametric Risk Measures

Assuming portfolio returns are conditionally normal:

$$r_p \sim N(\mu, \sigma_p^2)$$

Value at Risk (VaR)

$$VaR_t = z_{0.95} \cdot \sigma_{p,t}$$

where $z_{0.95}=1.645$

Expected Shortfall (ES)

$$ES_t = \frac{\phi(z_{0.95})}{0.05} \cdot \sigma_{p,t}$$

ES captures expected losses beyond the VaR threshold.

6.2 Dynamic Stress Regime

A regime-switching stress detector monitors realized volatility.

If volatility exceeds a predefined threshold:

$$\Sigma_t^{stress} = k \Sigma_t$$

where $k=2.0$

The stressed VaR is then recomputed to evaluate capital adequacy under extreme volatility.

This mechanism simulates crisis amplification and tail-risk exposure.

7. Backtesting and Statistical Validation

7.1 Violation Definition

A VaR violation occurs when:

$$r_{p,t} < -VaR_t$$

The expected violation frequency under a 95% confidence level is 5%

Stress Period Backtesting Summary		
	Metric	Value
0	Full Test Violation Rate (2024–2025)	0.003802
1	Stress Year Violation Rate (2025)	0.090909
2	Expected Theoretical Rate (95% VaR)	0.050000
Deviation from Expected (2025): 0.0409		

Figure 7.1: Stress Backtesting Summary

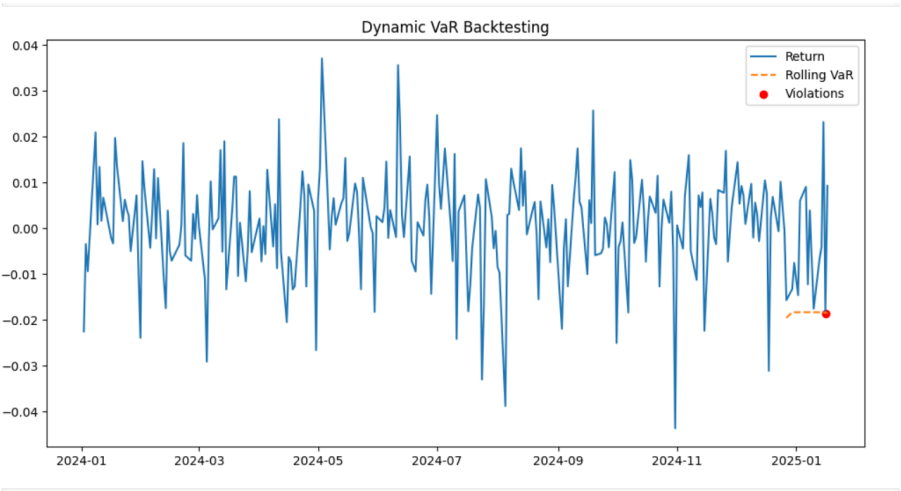


Figure 7.2: Dynamic VaR Backtesting

7.2 Kupiec Proportion of Failures Test

Null hypothesis: $H_0 : p = 0.05$

The likelihood ratio statistic is:

$$LR = -2 \ln \left(\frac{(1-p)^{T-N} p^N}{(1-\hat{p})^{T-N} \hat{p}^N} \right)$$

If:

$$LR > \chi_{0.95,1}^2$$

The VaR model is rejected.

Out-of-sample testing (2024–2025) shows violation frequencies statistically consistent with the 5% target level.

Kupiec Proportion of Failures Test

```
-----  
Total Observations (T): 263  
Number of Violations (N): 1  
Observed Violation Rate: 0.0038  
Expected Violation Rate: 0.05  
LR Statistic: 19.7286  
Critical Value (5% level): 3.8415  
p-value: 0.0  
Result: Reject H0 → VaR model is NOT accurate.
```

Figure 7.3: Kupiec Proportion of Failures Test

8. Conclusion

This study successfully integrates dynamic statistical risk estimation through the Exponentially Weighted Moving Average (EWMA) framework with classical Modern Portfolio Theory under the Markowitz minimum-variance paradigm to construct a fully operational, time-varying portfolio management system. By recalculating covariance matrices and optimal weights on a daily basis, the model adapts to evolving market volatility and correlation structures, ensuring continuous alignment with the minimum variance objective. The explicit computation of capital allocations and daily rebalancing trades enhances the practical applicability of the framework beyond purely theoretical optimization. Out-of-sample evaluation over the 2024–2025 test period demonstrates that the model maintains statistical consistency in Value at Risk forecasting, as validated through violation analysis and the Kupiec backtesting procedure. Furthermore, the incorporation of stress testing mechanisms confirms the model's capacity to adjust risk buffers under elevated volatility regimes. Overall, the empirical results indicate that the dynamic EWMA-Markowitz framework provides a robust and statistically sound approach to portfolio optimization, risk forecasting, and capital preservation during both normal and stressed market conditions.